Q1:JUN 2008 P2

5 **Fig. 5.1** shows two parallel metal plates separated by a distance d, with a potential difference, V between them. An oil drop of charge q and mass m is placed in the electric field between the plates.

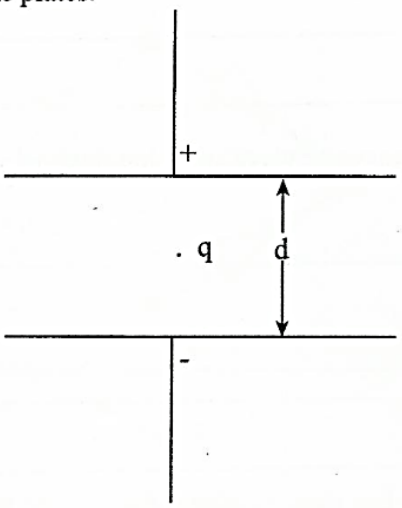


Fig. 5.1

(a) Define the term *electric field strength.*

____

____

____ [1]

(b) Write down an expression for

(i) the electric field strength E of this field and state its direction,

____ [2]

(ii) the electric force on the oil drop.

____

____ [1]

(c) Show that the potential difference required to keep the oil drop stationary is given by

$$V = \frac{mgd}{q}.$$

____

____

____

____ [2]

Q2:JUN 2010 P2

9 **(a)** Define *electric field strength* and give its S.I unit.

____

____

____ [2]

(b) An electron is placed separately in a gravitational field, an electric field and a magnetic field.

Complete **Table 9.1** by sketching diagrams to show the forces acting on the electron, the electron velocity and the field lines for the different states of motion given.

Table 9.1

field	**electron moving to the right**	**stationary electron**
Magnetic field		
Electric field		
Gravitational field		

[6]

Q3:NOV 2011 P2

5 (a) (i) Define *electric field strength.*

____

____

(ii) Give an expression linking the electric field strength to potential difference.

____ [2]

(b) **Fig. 5.1** shows a diagram for the 'shuttling ball experiment'.

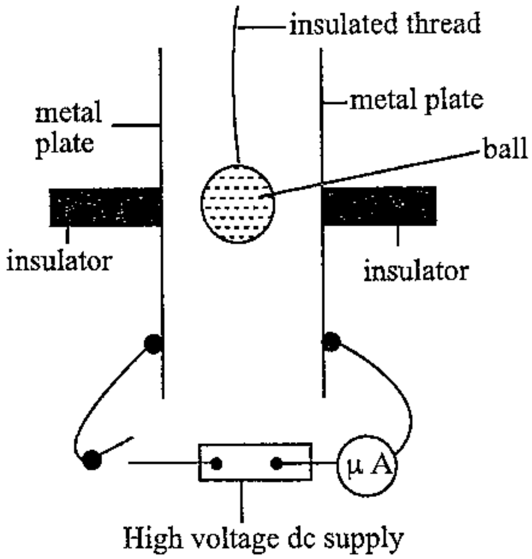


Fig. 5.1

The switch is now closed. Explain what happens to

(i) the metal plates,

____

____

(ii) the ball.

____

____

Q4:JUN 2013 P2

5 Electric forces can either be attractive or repulsive, while gravitational forces are always attractive. Consider an electron of charge $-e$ and a proton of charge $+e$ in an atom separated by a distance r.

(a) Write down the expression for the

(i) electric force, F_E, between the electron and the proton,

____

____

(ii) gravitational force, F_g, between the electron and the proton.

____

____

____

[2]

(b) State with a reason whether F_E in **(a) (i)** is attractive or repulsive.

____

____ [2]

(c) Using your answers in **(a)** and the appropriate numerical data, find the ratio of the magnitude of F_E to F_g.

[3]

Q5:NOV 2014 P2

5 **(a)** Define

(i) *electric field strength,*

____

____

____

(ii) *electric potential.*

____

____

____

[3]

(b) Two electric charges Q_1 and Q_2 of charges +10 nC and −5 nC respectively are placed 50 mm apart.

Calculate the combined electric field strength, E, at their mid-point.

E = ____________________ [3]

Q6:NOV 2015 P2

5 (a) Define *electric potential.*

____

____ [2]

(b) Equal and opposite $5\,\mu C$ charges are situated at the vertices X and Y of an equilateral triangle XYZ where XY = 10 cm.

Calculate the

(i) magnitude of the electric field strength at Z,

field strength = ______________________

(ii) electric potential at Z.

electric potential = ______________________ [4]

(c) Describe how dust can be extracted electrostatically from a chimney of a factory.

____

____

____

____ [3]

Q7:NOV 2016 P2

4 (a) Define *electric field strength.*

____ [1]

(b) An oil drop of weight of 3.04×10^{-14} N is held stationary between metal plates 4.0 mm apart when the potential difference applied is 380 V.

(i) Calculate the

1. electric field strength, E, between the plates,

$E =$ ____________________

2. magnitude of the charge on the oil drop.

charge = ______________

(ii) Hence show that the charge is quantized.

[6]

Q8:JUN 2017 P2

5 **(a)** Define

(i) *electric field strength,*

____

____

(ii) *electric potential.*

____

____ [2]

(b) An electron at rest, A, is moved in a vacuum by an electric field to B as shown in **Fig. 5.1**.

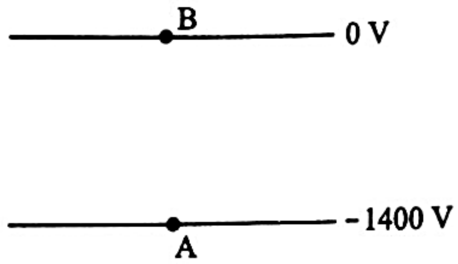


Fig. 5.1

(i) Show, on **Fig. 5.1**, the direction of the electric field.

(ii) Determine the

1. gain in electric potential by the electron,

gain in electric potential = ____________________

2. loss in electric potential energy by the electron,

3. velocity of the electron at B.

Q9:JUN 2018 P2

4 (a) (i) State Coulomb's law for the force between two point charges in a vacuum.

____

____

(ii) Explain any **two** ways in which Coulomb's law is similar to Newton's law of Gravitation for two point masses.

1.____

____

2.____

____ [3]

(b) (i) Estimate the distance between two protons in a nucleus.

____

(ii) Calculate the

1. Coulomb force between two protons,

Coulomb force = ______________________

2. gravitational force between the two protons.

(iii) Explain why protons inside the nucleus are bound to each other.

____

____

____ [6]

Q10:NOV 2003 P3

(a) Define the *electric field strength* and *electric potential* at a point. How are they related? [3]

(b) A beam of electrons travelling at 1.25×10^7 ms^{-1} enters a uniform electric field normally. The field is created by two parallel plates of length 60 mm, separated by a distance 25 mm. The upper plate is maintained at +50 V and the lower at −50 V as shown in **Fig. 6.1.**

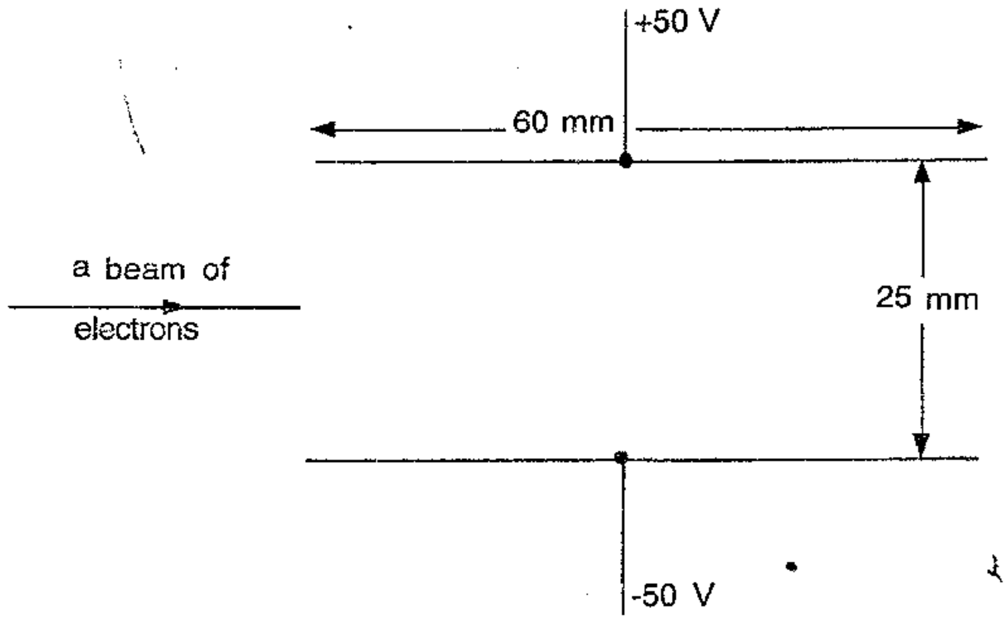


Fig. 6.1

Calculate

(i) the electric field strength between the plates,

(ii) the electric force on each electron,

(iii) the speed of the electrons as they leave the plates,

(iv) the gain in kinetic energy for each electron.

[11]

(c) Copy **Fig. 6.1** and complete the path of the beam between the plates and beyond. [1]

Q11:NOV 2018 P3

3 (a) (i) Define the terms

1. *electric field strength,*

2. *electric potential.*

(ii) State how electric field strength is related to electric potential.

(iii) Fig.3.1 shows a charged oil drop held stationary between two parallel horizontal plates.

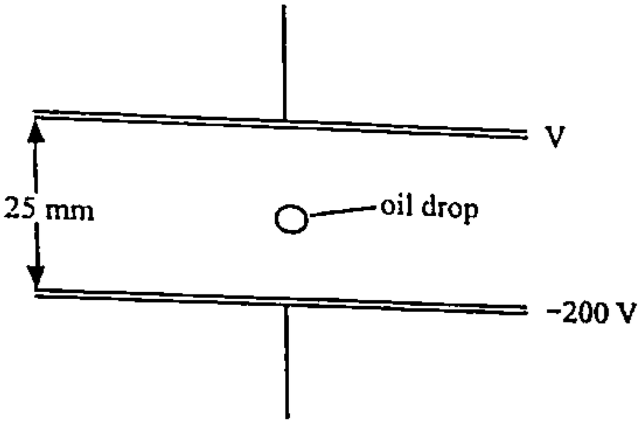


Fig.3.1

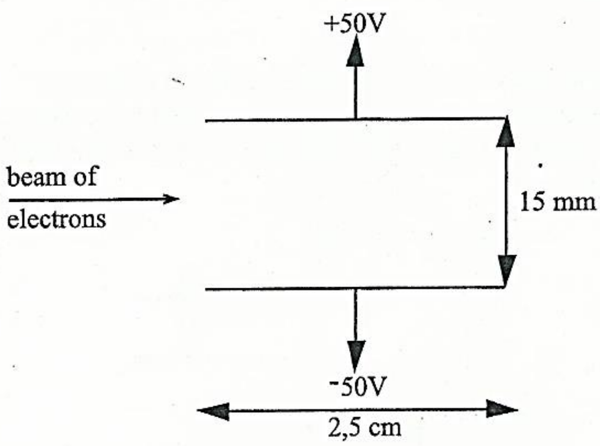
The oil drop has a mass of 5.0 mg and a charge of 64 μC.

Determine the value of the potential, V, on the top plate. **[7]**

Q12:NOV 2010 P5

3 (a) Define electric field strength and state its unit. [2]

(b) **Fig. 3.1** shows **two** parallel plates each of length 2.5 *cm* and a distance 15 *mm* apart. The upper plate is maintained at a potential of 50 *V* and the lower plate at a potential of $-50\ V$.



Q13:JUN 2012 P5

1 (b) (i) State and explain two ways by which a body can be charged.

(ii) Use Coulomb's Law to show that the electric field strength, E, at a distance , r, from a point charge, q, is given by

$$E = \frac{q}{4\pi\varepsilon_0 r^2}.$$

(iii) Two point charges, q_1 and q_2, are 50.0 cm apart. q_1 is 2.5×10^{-9} C and q_2 is -2.5×10^{-9} C. (See **Fig.1.1**)

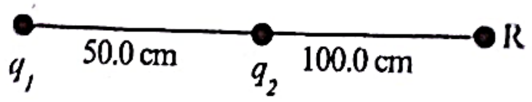


Fig.1.1

Determine the field strength at point R, 100.0 cm from q_2. [8]

(N.B. The diagram is not to scale).

4 (a) Define *electric field strength*. [1]

(b) Fig. 4.1 shows an electron getting into a uniform electric field, E, between two parallel plates of length, 200 cm.

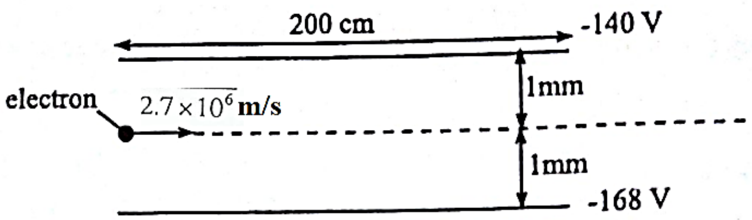


Fig. 4.1.

(i) Determine the magnitude of the electric field strength, E.

(ii) Calculate the electric force on the electron.

(iii) On a sketch diagram, label all the forces acting on the electron.

(iv) Deduce whether the electron will leave the field or not. Hence complete the path of the electron.

Q14:JUN 2014 P5

1 (a) (i) Define *electric potential at a point.*

(ii) Three point charges **A**, **B** and **C** are situated in air as shown in **Fig. 1.1**.

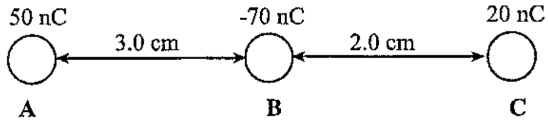


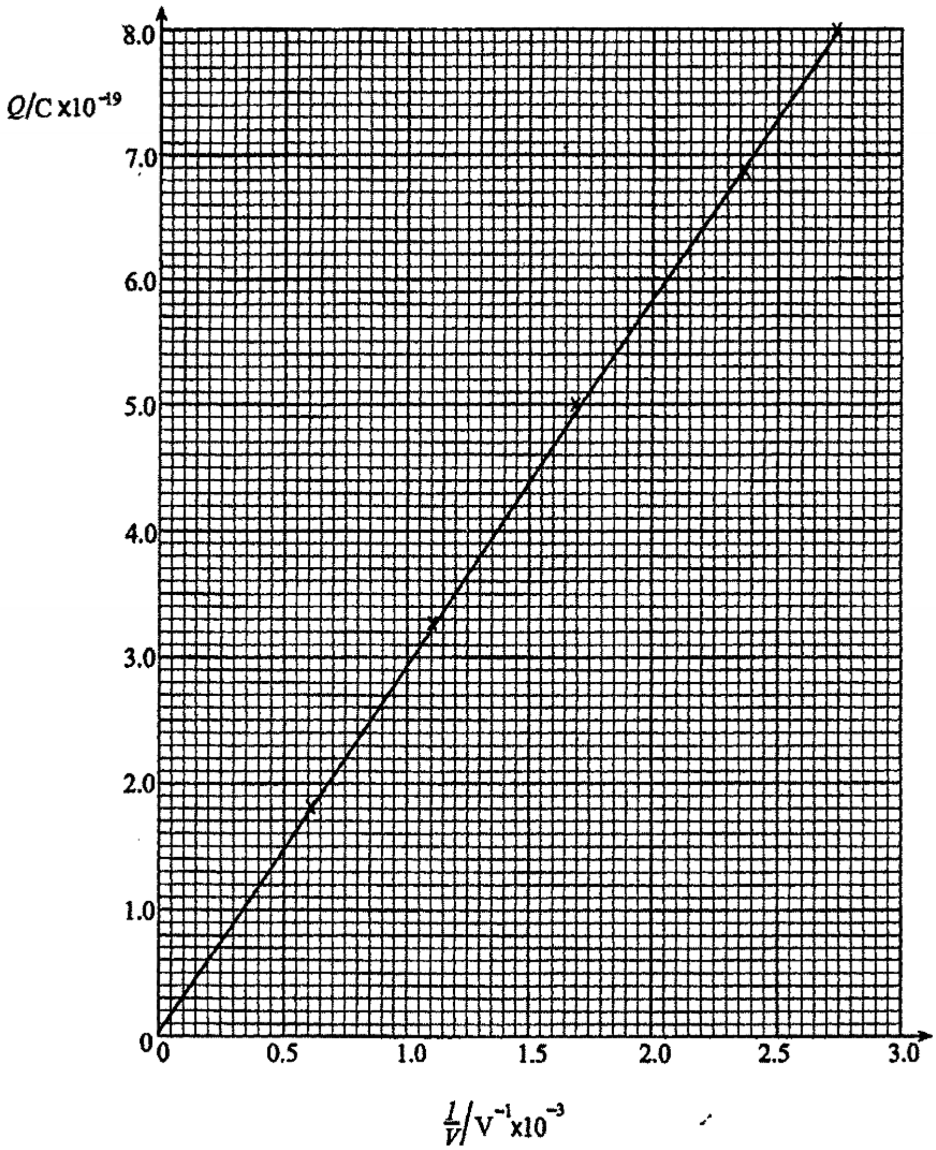
Fig. 1.1

Calculate the resultant force acting on **C**.

[4]

Q15:JUN 2016 P5

2 (a) **An oil drop, of mass *m*, is given a charge, *Q*, and is allowed to enter a uniform electric field. It is held stationary between two horizontal parallel plates by adjusting the electric field such that a potential difference, *V*, is between the plates. The charge, *Q*, is changed and the corresponding potential difference, *V*, to keep the drop stationary is obtained. Fig. 2.1 shows the results.**



(i) Show that Q is related to V and m by the expression

$$Q = \frac{0.196\,m}{V},$$

where the upthrust is negligible and the separation between the plates is 20.0 mm.

(ii) Use **Fig. 2.1** to determine the mass of the oil drop.

[7]

Q16:JUN 2016 P5

4 (a) Define *electric field strength.* [1]

(b) (i) Sketch electric field lines of a unit negative charge.

(ii) Mark on the sketch in (i) any two points A and B at the same potential.

[2]

(c) **Fig. 4.1** shows two small charged spheres, X and Y that are 1.0 m apart in air.

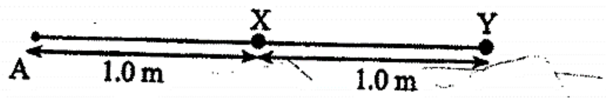


Fig. 4.1

Find the electric field strength, at point A, due to X and Y carrying charges of 1.0×10^{-8}C and -1.0×10^{-8}C respectively. [5]

(d) Compare, quantitatively, the *electrostatic* and *gravitational* forces of a proton at a distance, r, from an electron. [4]